

1.11 Three E-ACs And A Transistor

1946

The ENIAC (Electronic Numerical Integrator and Computer) was one of the first (a) totally-electronic, (b) valve-driven, and (c) digital computers. Development finished in 1946 at the Ballistic Research Laboratory in the USL it was designed by J Presper Eckert and John Mauchly. (Eckert was the brains, and Mauchly was the engineer, roughly speaking.) Yet another first, this one was widely recognised as the first Universal Electronic Computer. It weighed 30 tonnes, and could handle 1,00,000 calculations per second.

It was not used for entirely benign purposes: the numbers it crunched were those of ballistic trajectories and tests of hydrogen bomb theories.

1947

This was it—the invention of the century, indeed, no invention has surpassed this one to date. On 16 December, 1947, at The Bell Laboratories, William Shockley (first credit), John Bardeen, and Walter Brattain invented the transistor.

We've explained in *Digit* how this three-legged marvel does its job, and how the silicon in it is being replaced by carbon nanotubes; suffice it now to say that much of your life depends on it!

William Bradford Shockley was a controversial figure: he was racist, he wanted all the fame (around the transistor) to himself,



John Mauchly (far left) and J Presper Eckert (far right) with Maj. Gen. Barnes, head of research and development for the army ordnance, reviewing the ENIAC maintenance records. Yes, computers were a big deal back then!